

Zain Sarwar

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RESEARCH INTERESTS

Memory efficient reasoning, Mixture of Experts, Inference optimization for LLMs, Neural Memory architectures

EDUCATION

University of Chicago

Chicago, IL

PhD in Computer Science

Aug 2025 (Expected)

GPA: 3.91/4.0

Relevant Coursework: Deep Learning, Machine Learning, Operating Systems, NLP, Algorithms & Distributed systems

PROFESSIONAL EXPERIENCE

Capital One

New York, NY

Applied research intern - LLM Core & Agentic AI

June 2025-Sept 2025

Project : Structured Reasoning With Thought Pruning

- Developed a method for enabling reasoning based LLMs to reason in structured text to enable preservation of high level information pathways
- This method improves reasoning performance of SOTA reasoning models like Qwen & Phi by 4-5%
- Improved memory and compute efficiency by pruning traces on-the-fly with minimal performance degradation

Capital One

New York, NY

Applied research intern - LLM Pretraining

June 2024 – Sept 2024

Project : Improving Mixture of Experts with routed structured matrices

- Developed a method for augmenting experts in MoEs with structured matrices using hierarchical routing
- This method provides an alternative scaling strategy for MoE based LLMs which has a more efficient inference profile and outperforms a parameter and FLOP matched standard MoE in large scale pretraining.
- Evaluated method by pretraining 125M, 410M and 1.2B parameter models on multiple pretraining datasets, routing schemes, total activated parameters and learning rate schedules

University of Chicago

Chicago, IL

Research Assistant

September 2021 – Present

Ongoing projects

- Designing stack based memory architectures for neural networks

Past projects

- Invented a first-of-its-kind safety analysis tool for LLMs which can detect the model's tendency to hallucinate sensitive information by crafting semantically meaningful prompts using a retrieval mechanism
- Engineered state-of-the-art techniques for detecting text generated from Large Language models using graph neural networks that can be used to detect LLM generated fake news and hate speech

SELECTED PUBLICATIONS

[StructMoE: Structured Mixture of Experts Using Low Rank Experts](#)

Zain Sarwar, Ashwinee Panda, Benjamin Thérien, Supriyo Chakraborty et al.

NEURIPS 2024 ENLSP Workshop, Vancouver, Canada

[Continual Pre-training of MoEs: How robust is your router?](#)

Benjamin Thérien, Charles-Étienne Joseph, Zain Sarwar et al.

[Dense Backpropagation Improves Routing for Sparsely-Gated Mixture-of-Experts](#)

Ashwinee Panda, Varsal Baherwani, Zain Sarwar, Benjamin Thérien, Supriyo Chakraborty et al.

NEURIPS 2024 Compression Workshop, Vancouver, Canada

[Deepfake Text Detection: Limitations and Opportunities](#)

Jiameng Pu, Zain Sarwar, Sifat Muhammad Abdullah, Abdullah Rehman, Mobin Javed, and Bimal Viswanath

IEEE S&P (Oakland) 2023

TECHNICAL SKILLS

Languages: Python, Triton, C, C++, Java, JavaScript, Go, Haskell, Matlab, SQL

Libraries: PyTorch, DeepSpeed, Megatron-LM, TensorFlow, OpenCV, scikit-learn, pandas, NumPy, Selenium

Frameworks: Angular, React, Git, Docker, Flask, Node.js, Vue